Final exam preparation exercises.

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0. Let $A = \begin{bmatrix} -1 & -2 & -3 & 2 \\ 2 & -1 & 1 & 3 \\ 7 & 4 & 11 & 2 \end{bmatrix} = \begin{bmatrix} — & a_{r_1} & — \\ — & a_{r_2} & — \\ — & a_{r_3} & — \end{bmatrix} = \begin{bmatrix} | & | & | & | \\ a_{c_1} & a_{c_2} & a_{c_3} & a_{c_4} \\ | & | & | & | \end{bmatrix}$, answer the following questions.

0.1 Solve $Ax = 0$.

0.2 $\text{rank}(A) =$ _____, $\text{nullity}(A) =$ _____

0.3 What is the relation between 0.1 and 0.2?

0.4 Solve $A^T x = 0$. (Assuming you know 0.1 – 0.2, can this be done instantly, how?)

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0.5 If $f(x) = Ax$, is f onto, one-to-one, either, or neither?

0.6 If $g(x) = A^T x$, is g onto, one-to-one, either, or neither?

0.7 Is $\left\{a_{r_1}, a_{r_2}, a_{r_3}\right\}$ linearly independent?

0.8 Is $\left\{a_{c_1}, a_{c_2}, a_{c_3}, a_{c_4}\right\}$ linearly independent?

0.9 What is the relation between 0.7/0.8 and 0.2?

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0.10 Find a basis that spans the same set as the span of the set in 0.7

Which of the four fundamental subspaces does this basis represent?

0.11 Find a basis that spans the same set as the span of the set in 0.8

Which of the four fundamental subspaces does this basis represent?

0.12 Let's say that we add the fourth row to A and make this matrix square, called it A_{sq} .

Is $\det\left(A_{\text{sq}}\right)$ deterministic? In other words, can $\det\left(A_{\text{sq}}\right)$ be described for any fourth row of A_{sq} ?

From 0.1 – 0.12, be noted that all sub questions related to one another nicely.

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1. Provide short answer to each question.

1.1 Let A be a matrix that represents a linear map whose domain is $\mathbb{R}^5$ and range is $\mathbb{R}^3$.

Now consider a pair of $\left(\text{rank}\left(A\right), \text{nullity}\left(A\right)\right)$. How many different pairs are possible for this A.

1.2 Let $a_1 = \left(1,2,3\right)$ and $a_2 = \left(4,-4,1\right)$, write a vector with minimum length that is a member of $\left\{a_1, a_2\right\}^{\perp}$.

1.3 Consider a matrix $A = \begin{bmatrix} 2 & 1 & a \\ 0 & -3 & 6 \\ 0 & 0 & b \end{bmatrix}$, give one example value of both a and b that makes $\text{nullity}\left(A\right) = 1$.

1.4 A mapping $f : \mathbb{R}^2 \to \mathbb{R}^3$, is given by the rule $F\left(x,y\right) = \left(x,y,y\right)$

Is f one-to-one, onto, expand, or contract?

1.5 Let A be an $m \times n$ matrix such that the nullity of AA^T is c.

Find $\text{rank}\left(A\right) + \dim\left(\text{row}\left(A\right)\right) + \dim\left(\text{N}\left(A\right)\right)$ in terms of m, n, and c.

1.6 What are the number of columns and row of a smallest matrix A such that $\text{col}\left(A\right) = \text{span}\left(\left\{\left[-1,0,1,0\right]^T\right\}\right)$.

1.7 Write a set P such that $\text{span}\left(P\right) = S \cap S^{\perp}$ where $S = \left\{\left[1,0,0,0\right]^T\right\}^{\perp}$.

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1.8 Consider a set $S = \left\{ \begin{bmatrix} 2 \\ 6 \end{bmatrix}, \begin{bmatrix} 3 \\ -5 \end{bmatrix}, \begin{bmatrix} -3 \\ -9 \end{bmatrix}, \begin{bmatrix} -6 \\ 10 \end{bmatrix} \right\}$. How many subsets of S are linearly independent?

Noted that there are 16 possible subsets of S.

1.9 Let $a_1 = \left[-1, 0, 0\right]^T$, $a_2 = \left[0, 1, 0\right]^T$ and $a_3 = \left[x, y, z\right]^T$.

If $\left\{a_1, a_2, a_3\right\}$ is an orthonormal basis of $\mathbb{R}^3$ and $x + y + z \neq 1$. Find a_3.

2. Consider a matrix A where some entries are missing, $A = \begin{bmatrix} -4 & 2 & -2 & 6 \\ 3 & -1 & 1 & 0 \\ 8 & a & 3 & b \end{bmatrix}$ for some real numbers a and b.

We know that the basis of $\operatorname{row}\left(A\right)$ contains exactly 2 vectors.

Calculate the following sets with an explanation, i.e., describe why calculation gives the required set.

2.1 Basis of $\operatorname{row}\left(A\right)$.

2.2 Smallest set S such that $S^{\perp} = \operatorname{nul}\left(A\right)$

2.3 Smallest set T such that $\operatorname{span}\left(T\right) = \operatorname{col}\left(A\right)$

2.4 Smallest set U such that $\operatorname{span}\left(U\right) = \operatorname{nul}\left(A^T\right)$

3. Let $h:\mathbb{R}^3 \to \mathbb{R}^2$ be a linear map and A be a matrix that represents h. Given that,

- $h\left(\begin{bmatrix}1\\0\\0\end{bmatrix}\right)=\begin{bmatrix}-2\\4\end{bmatrix}$.
- $\begin{bmatrix}4\\2\end{bmatrix}$ is in the left nullspace of A.
- $B=\left\{b_1,b_2,\begin{bmatrix}1\\0\\0\end{bmatrix}\right\}$ and $D=\{d_1,d_2\}$ are orthogonal (not necessarily orthonormal) bases of $\mathbb{R}^3$ and $\mathbb{R}^2$ respectively.
- $\mathrm{Rep}_{B,D}(h)=\begin{bmatrix}-2 & 5 & 1\\0 & 0 & 0\end{bmatrix}$

Find A, B and D that together satisfy all the information above. Explanation is necessary or no point will be awarded.

*Additional question; is the condition "$\begin{bmatrix}4\\2\end{bmatrix}$ is in the left nullspace of A." needed, why?

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4. Given matrices P, Q, and X in which X is invertible and $P = XQ$. Let $X = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 \\ 4 & 5 & 6 & 0 \\ 7 & 8 & 9 & 10 \end{bmatrix}$, $P = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 5 & 7 & 9 \\ 4 & 9 & 13 & 17 \\ 5 & 12 & 17 & 22 \end{bmatrix}$

and $x = \begin{bmatrix} 10 \\ 1 \\ -\frac{1}{3} \\ -\frac{19}{10} \end{bmatrix}$ to be one of the possible solutions to $Qx = b$. Find all solutions to $Qx = b$.

5. Shows that for any invertible matrix $A_{m\times m}$ and $B_{m\times n}$ for all positive integers m and n.

5.1 $\operatorname{nul}\left(AB\right) = \operatorname{nul}\left(B\right)$

5.2 $\operatorname{rank}\left(AB\right) = \operatorname{rank}\left(B\right)$

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6. Provide short answer to each question.

6.1 Let $a,b \in \{-2,-1,0,1,2\}$ and $A = \begin{bmatrix} 1 & a & b \\ 0 & a & a \\ 0 & 0 & b \end{bmatrix}$. How many possible (a,b) are there which makes A diagonalizable.

6.2 If a matrix A can be diagonalized to $A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 10 & 0 \\ 0 & -4 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}^T$ and has UDV^T as its SVD.

Find one possible U, D and V that together form the SVD of A.

6.3 Find eigenvalue(s) along with their algebraic and geometric multiplicity of $\begin{bmatrix} -1 & -1 & 4 & 0 \\ 0 & 4 & 7 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$.

6.4 Let $A = \begin{bmatrix} 1 & 1 & 5 \\ 0 & 3 & 2 \\ 0 & 0 & 2 \end{bmatrix}$. Find the largest eigenvalue of A^3 along with its eigenvector.

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For 6.5 – 6.7, Let A be a 2×2 matrix such that $A\begin{bmatrix}8\\6\end{bmatrix}=\begin{bmatrix}18\\24\end{bmatrix}$, and $U\Sigma V^T$ be an SVD of A such that the first column of U is $\begin{bmatrix}0.6\\0.8\end{bmatrix}$ and one of the singular values of A is 2.

6.5 Find the first column of V.

6.6 Find (all possible) determinant of A.

6.7 How many different matrix A are possible?

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7. Find the characteristic polynomial, all the eigenvalues and its corresponding eigenvectors for $A = \begin{bmatrix} 5 & 2 & 0 \\ 2 & 5 & 0 \\ -3 & 4 & 6 \end{bmatrix}$.

   Explanation is necessary or no point will be awarded.

8. Suppose A has the eigenvalues $\lambda_1 = 1, \lambda_2 = 0,$ and $\lambda_3 = -1$ with eigenvectors $x_1,\ x_2,$ and x_3 respectively.

Given the matrix S as follows, $S = \begin{bmatrix} x_1 & | & x_2 & | & x_3 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$.

Explanation for each question is necessary or no point will be awarded.

8.1 Compute the eigenvalues along with its eigenvectors of $B = A^{2565} + I$

8.2 Is B invertible?

9. Let A be a matrix with linearly independent columns and consider the projection matrix $P = A\left(A^T A\right)^{-1} A^T$.

   (As you have remembered, Pb is a projection of b on $\mathrm{col}\left(A\right)$.) What are the possible eigenvalues for P?

   Explanation is necessary or no point will be awarded.

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10. Consider the diagonalization of a matrix $A = STS^{-1}$ where $S = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 2 & 0 & 0 \\ 7 & -1 & 4 & 0 \\ -5 & 3 & -2 & 7 \end{bmatrix}$ and $T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix}$.

If $M = A^4 + 2A^3 - 7A^2 - 8A + 12I$, find $\operatorname{rank}\left(M\right)$. Explanation is necessary or no point will be awarded.